

- 8 (a)** Fig. 8.1 shows part of the orbit of a space capsule around the Earth.

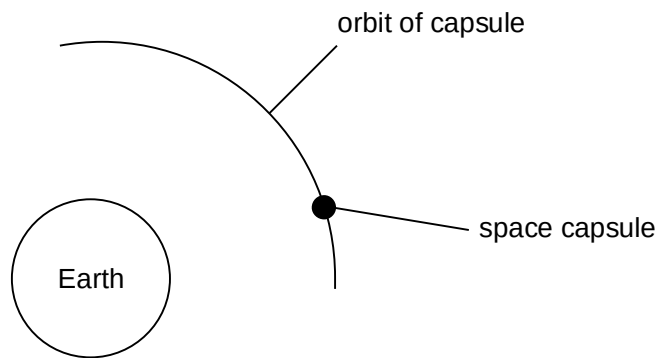


Fig. 8.1

- (i)** State Newton's law of gravitation.

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 [2]

- (ii)** An astronaut in the space capsule experiences a sense of 'weightlessness'. Explain why the astronaut has this experience in the capsule.

.....

 [2]

- (iii) The space capsule of mass m orbits the Earth of mass M_E . The gravitational potential energy, U of the space capsule is expressed as

$$U = - \frac{GM_E m}{r}$$

where r is the radius of orbit.

By deriving the kinetic energy of the space capsule in orbit, obtain an expression for total energy, E_T of the satellite in terms of G , M_E , m and r .

[3]

- (b) The Earth may be assumed to be a uniform sphere of radius 6.38×10^6 m. A satellite of mass 2400 kg is projected from the Earth's surface and placed in a geostationary orbit at a distance of 4.23×10^7 m from the centre of the Earth.

- (i) State what is meant by *geostationary*.

.....

 [1]

- (ii) State the direction of orbit of the satellite around the Earth's axis.

..... [1]

- (iii) Explain why the satellite must be above the equator.

.....

 [3]

(iv) Calculate the speed of the satellite placed in this orbit.

speed = m s^{-1} [2]

(v) Using the answer obtained in (b)(iv), calculate the change in gravitational potential energy of the satellite when it is launched into a geostationary orbit from the Earth's surface.

change in gravitational potential energy = J [3]

(vi) Hence, determine the minimum speed required to project the satellite from the Earth's surface to reach this orbit.

minimum speed = m s^{-1} [3]

9 (a) By reference to the photoelectric effect state what is meant by the *work function*